

CZESTOCHOWA, Poland

WMO: 125500

Lat: 50.8118N

Lon: 19.0915E

Elev: 968

StdP: 14.19

Time Zone: 1.00 (EUC)

Period: 00-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	7.0	12.0	1.5	6.2	8.9	5.8	7.7	14.0	18.6	34.8	16.2	34.2	5.0	90	0.488

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	17.3	86.1	67.2	82.7	65.6	79.8	64.4	69.3	81.5	67.8	78.9	66.3	76.4	5.1	200

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
65.4	97.4	73.4	63.9	92.1	71.9	62.5	87.7	70.6	33.9	82.4	32.6	79.4	31.5	76.6	74.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4)	15.1	12.9	11.3	DB	2.2	91.8	7.9	2.9	-3.4	93.9	-8.1	95.7	-12.5	97.3	-18.2	99.5
(5)				WB	1.5	72.3	7.7	1.3	-4.0	73.3	-8.5	74.1	-12.8	74.8	-18.4	75.8

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	48.5	28.9	30.6	37.9	49.3	56.9	63.7	67.1	66.0	57.9	48.9	40.9	32.8
	DBStd	15.42	9.40	9.06	8.52	7.84	7.28	6.52	6.16	5.77	6.28	7.77	7.48	7.93
	HDD50	2645	654	543	380	105	20	0	0	0	8	118	283	534
	HDD65	6314	1119	963	839	472	265	100	51	58	227	498	723	999
	CDD50	2103	0	0	6	85	235	412	530	497	244	85	9	0
	CDD65	299	0	0	0	2	16	62	116	90	12	1	0	0
	CDH74	2470	0	0	0	29	161	515	946	710	107	2	0	0
	CDH80	616	0	0	0	3	23	122	267	185	16	0	0	0
Wind	WSAvg	5.6	6.3	6.2	6.6	6.3	5.5	5.2	4.9	4.6	4.8	5.1	5.7	6.3
Precipitation	PrecAvg	25.6	1.5	1.3	1.7	1.7	2.9	3.1	3.7	2.4	2.4	1.7	1.8	1.4
	PrecMax	34.9	3.5	3.0	3.9	3.8	8.2	6.9	9.8	4.9	5.8	3.7	3.2	4.3
	PrecMin	17.8	0.3	0.3	0.6	0.1	1.1	1.0	1.5	0.9	0.5	0.2	0.0	0.4
	PrecStd	4.0	0.7	0.7	0.8	0.8	1.5	1.4	1.9	1.2	1.2	1.1	0.7	0.8
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	50.1	54.3	64.7	78.1	83.1	88.7	90.5	89.9	82.3	72.9	61.7	53.4
		MCWB	45.2	45.4	50.3	58.7	63.1	68.7	68.8	68.8	62.8	59.7	53.1	47.6
	2%	DB	46.2	49.5	59.1	71.5	78.7	83.4	86.6	84.6	77.1	68.5	57.5	48.4
		MCWB	42.8	43.2	47.6	54.4	61.7	66.4	67.3	67.0	62.7	58.2	51.0	44.9
	5%	DB	43.3	45.8	54.9	67.0	74.5	79.6	82.7	81.2	73.1	64.4	54.0	45.4
		MCWB	40.4	41.1	45.8	52.5	60.0	64.3	66.0	65.9	60.6	56.7	49.5	42.6
	10%	DB	40.7	42.1	50.8	63.1	70.6	76.3	79.4	77.8	69.5	60.6	50.9	42.7
		MCWB	38.5	38.6	44.1	50.8	58.0	62.7	64.6	64.3	59.0	54.6	47.6	40.8
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	46.4	47.0	51.7	59.1	65.6	70.6	71.8	70.8	65.7	61.6	54.1	48.4
		MCDB	48.3	52.3	61.6	75.9	78.8	83.9	85.7	83.0	77.7	69.4	59.2	52.2
	2%	WB	43.1	44.0	49.0	56.4	63.2	68.1	69.5	68.8	63.6	59.2	52.1	45.2
		MCDB	45.8	48.3	57.0	68.7	74.8	79.7	81.3	81.3	73.4	66.1	56.5	48.0
	5%	WB	40.7	41.4	46.7	54.2	61.3	66.3	68.0	67.3	61.9	57.1	49.9	42.9
		MCDB	42.9	45.4	53.2	64.6	71.6	76.9	78.7	78.4	70.4	63.3	53.5	45.0
	10%	WB	38.5	39.0	44.5	51.9	59.4	64.3	66.5	65.7	60.2	55.0	47.7	40.8
		MCDB	40.5	41.8	50.1	61.3	69.0	73.7	76.0	75.1	67.9	60.2	50.6	42.5
Mean Daily Temperature Range	5% DB	MDBR	8.5	10.2	13.3	17.1	16.9	17.0	17.3	17.3	15.9	13.6	9.7	7.9
		MCDBR	9.7	15.0	20.7	23.3	23.0	22.3	22.2	22.5	21.5	19.1	14.9	10.2
		MCWBR	7.6	9.8	12.0	10.9	9.6	9.1	7.6	8.7	9.6	10.3	9.6	7.9
	5% WB	MCDBR	9.3	13.4	18.6	21.3	20.9	20.2	19.5	20.0	19.1	18.0	13.6	9.8
		MCWBR	7.6	9.5	11.4	10.6	9.7	9.5	7.7	8.4	9.5	10.4	9.3	8.1
Clear-Sky Solar Irradiance	taub	0.316	0.341	0.380	0.432	0.426	0.422	0.445	0.434	0.401	0.370	0.347	0.313	
	taud	2.275	2.262	2.214	2.132	2.202	2.244	2.192	2.225	2.287	2.340	2.342	2.327	
	Ebn at Noon	220	246	257	255	264	266	256	252	247	233	207	203	
	Edh at Noon	22	29	36	44	43	42	43	40	34	27	21	18	
All-Sky Solar Radiation	RadAvg	251	482	848	1298	1597	1682	1598	1432	1025	599	287	194	
	RadStd	46	84	117	167	184	169	197	119	143	87	39	25	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	N/A	N/A	N/A	N/A	+1.76	+1.40	N/A	N/A	N/A	N/A
(47)	Regional (34 neighbors)	+0.90	N/A	N/A	+1.40	+0.59	N/A	N/A	-295	+121	+47

Nomenclature: See separate page